Linear Algebra & Matrix Applications

Engineering Application Problem Set

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1. Course Mapping & Syllabus Overview

This document links core concepts of Matrix Algebra—including Elementary Transformations, Systems of Linear Equations, Eigenvalues/Eigenvectors, and Diagonalization—directly to practical engineering domains such as Error-Correcting Codes (Digital Communications) and Loop Current Analysis (Electrical Network Theory).

Calculus Topic	Engineering Application Area	Key Mathematical Tool
Rank & Echelon Form	Error Detection & Correction Codes	Row Echelon Form (REF), Parity-Check Matrix rank
Direct Methods (Gauss Elimination)	Loop Current Circuit Analysis	Forward Elimination & Back Substitution
Iterative Methods (Gauss-Seidel)	Large-Scale Power Flow & Mesh Networks	Strictly Diagonally Dominant Iterations
Eigenvalues & Eigenvectors	Dynamic Systems & State Equations	Characteristic Equation $\det(A - \lambda I) = 0$
Diagonalization	Decoupling Network Equations	Modal Matrix P, $P^{-1}AP = D$
Elementary Transformations & Inverse	Network Topology & Mesh Analysis	Elementary Row Operations (ERO), $[A \mid I] \rightarrow [I \mid A^{-1}]$

Theoretical Concepts & Engineering Applications

Error Detection & Correction Codes

In digital electronics, **error detecting and correcting codes** are techniques used to maintain data integrity when binary data (0s and 1s) is transmitted over a noisy channel or stored in hardware. Electrical noise or interference can flip a bit (changing a 0 to a 1 or vice versa), causing corruption.

To protect the data, these codes add **redundant bits** (extra bits) to the original message before transmission. The relationship between data bits and redundant bits allows the receiver to determine if an error occurred and, in some cases, pinpoint exactly where it happened to fix it.

Error Detecting Codes

These codes only detect that an error has occurred; they cannot determine where the flipped bit is located. If an error is caught, the system typically requests a retransmission.

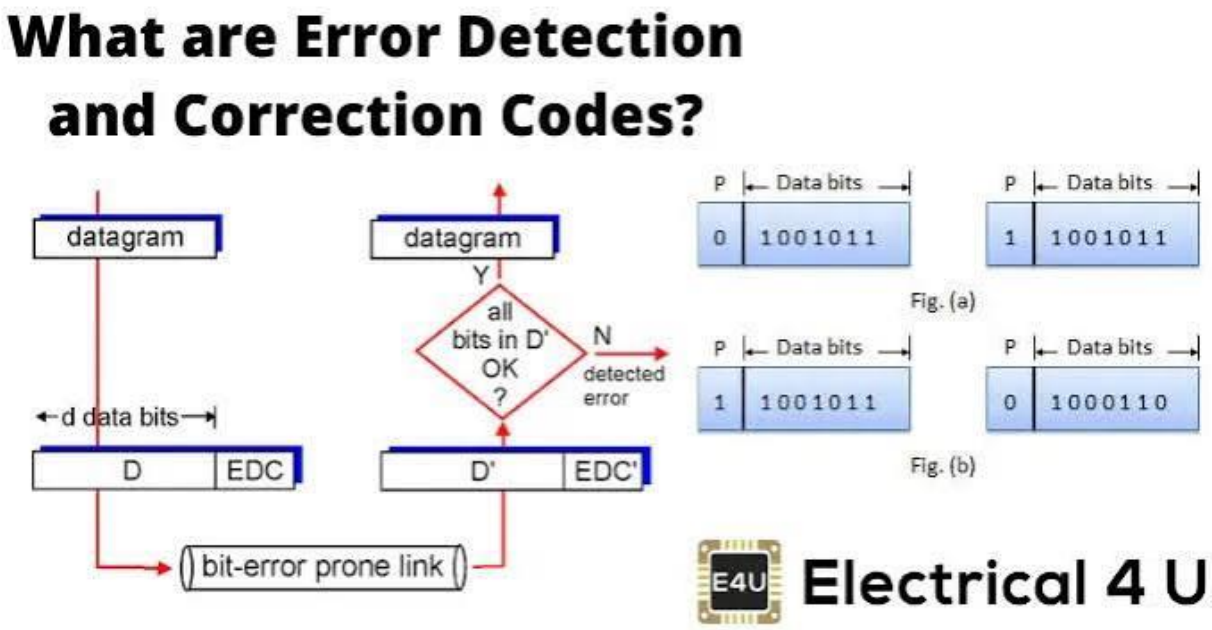
- **Parity Bit:** A single bit appended to a data block to ensure that the total number of 1s is either even or odd.
    - *Even Parity:* The parity bit is set to 0 or 1 so that the total count of 1s is **even**.
    - *Odd Parity:* The parity bit is set so that the total count of 1s is **odd**.
    - *Limitation:* It can only detect an **odd number of bit errors** (e.g., a 1-bit flip). If two bits flip simultaneously, the parity remains valid, and the error goes undetected.
- **Checksum:** The data is split into fixed-size segments, added together using 1's complement arithmetic, and the result is complemented to form a checksum. The receiver repeats the math; if the final result is not zero, an error is detected.
- **Cyclic Redundancy Check (CRC):** A highly reliable polynomial division method. The sender appends a sequence of bits (remainder of a binary division) to the data stream. The receiver performs the same division. If the remainder is non-zero, an error has occurred.

Error Correcting Codes (ECC)

These codes detect the presence of an error and pinpoint its precise bit location, allowing the receiver to automatically flip it back without needing retransmission. [

- **Hamming Code:** A widely used linear block code that inserts multiple parity bits at strategic, power-of-two positions (1,2,4, 8,…..) within the data.
    - The relationship between the data bits (n) and parity bits (k) must satisfy the formula:

$$2^k \geq n + k + 1$$

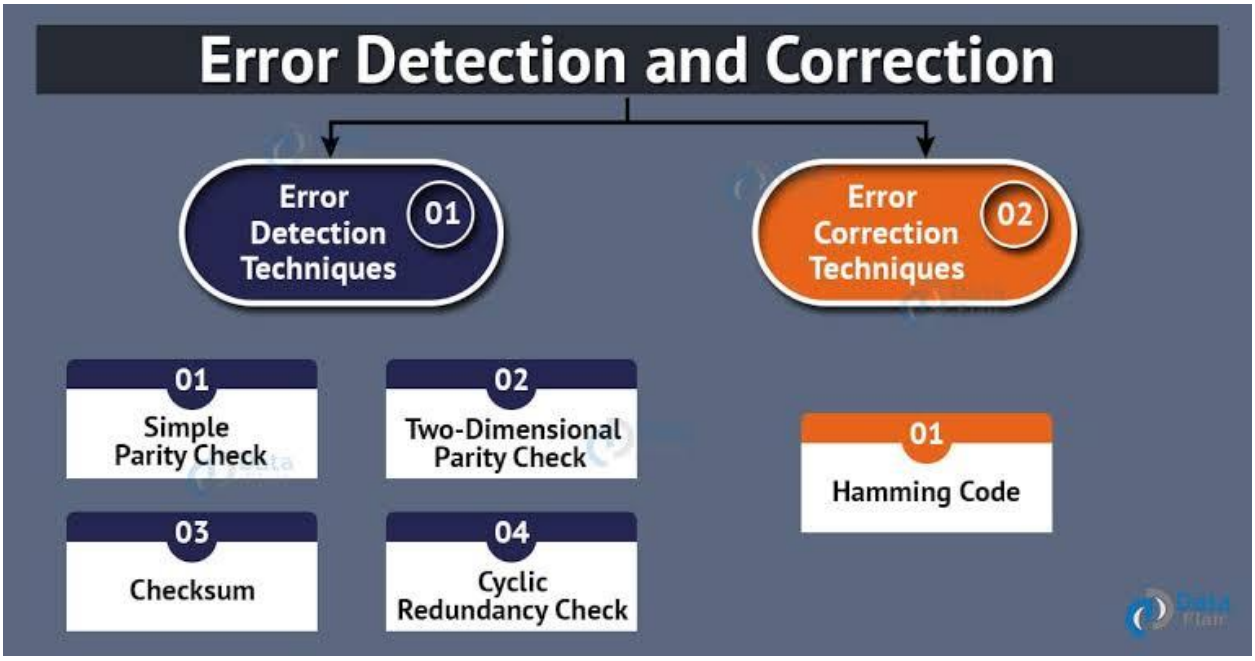


- The decimal value of the calculated check bits at the receiver side (called the **syndrome**) identifies the exact bit position of the error. Standard Hamming code can **correct single-bit errors** and **detect double-bit errors**.

- **Reed-Solomon Codes:** Non-binary cyclic codes that operate on blocks of data (symbols) rather than individual bits. They are exceptionally efficient at correcting **burst errors** (where consecutive multiple bits are corrupted) and are heavily used in CDs, DVDs, and QR codes. [1]

Technique	Type	Key Characteristic	Detection Capability	Correction Capability
Parity Check	Error Detection	Adds 1 extra bit per data word	1-bit errors	None
Cyclic Redundancy Check (CRC)	Error Detection	Polynomial division remainder	Burst errors	None

Technique	Type	Key Characteristic	Detection Capability	Correction Capability
Hamming Code	Error Detection & Correction	Uses parity bits at powers-of-two positions	2-bit errors	1-bit error
Reed-Solomon Code	Error Detection & Correction	Block-based non-binary code	Multiple burst errors	Multiple symbol errors



Current Loop Analysis

Current Loop Analysis (also called **Mesh/Loop Current Analysis**) is a systematic method used to determine the unknown currents flowing through the loops of an electrical circuit by applying **Kirchhoff's Voltage Law (KVL)** to each independent loop.

In this method:

1. A **loop current** is assumed for each independent loop.

2. KVL is applied around each loop.
3. The voltage drops across resistors are expressed using **Ohm's law**, $V = IR$.
4. The resulting simultaneous linear equations are solved to obtain the loop currents.

For example, for a two-loop circuit, the equations can be written in matrix form as

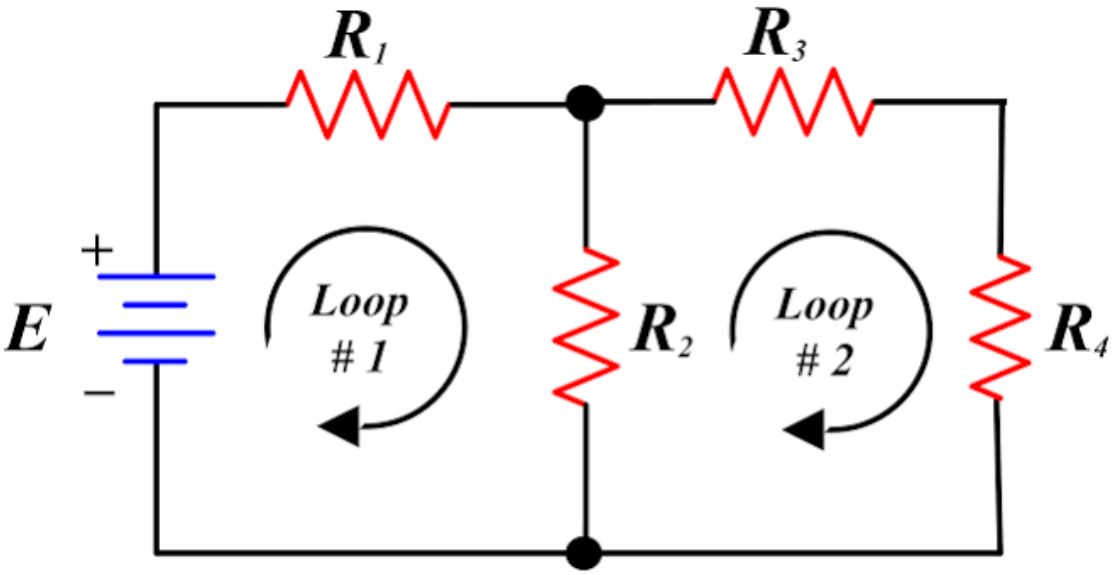
$$\begin{bmatrix} R_1 + R_3 & -R_3 \\ -R_3 & R_2 + R_3 \end{bmatrix} \begin{bmatrix} I_1 \\ I_2 \end{bmatrix} = \begin{bmatrix} V_1 \\ V_2 \end{bmatrix}.$$

Thus, **loop current analysis provides a direct application of matrix algebra to electrical circuit analysis**, where the unknowns are the loop currents and the coefficients are determined by circuit resistances.

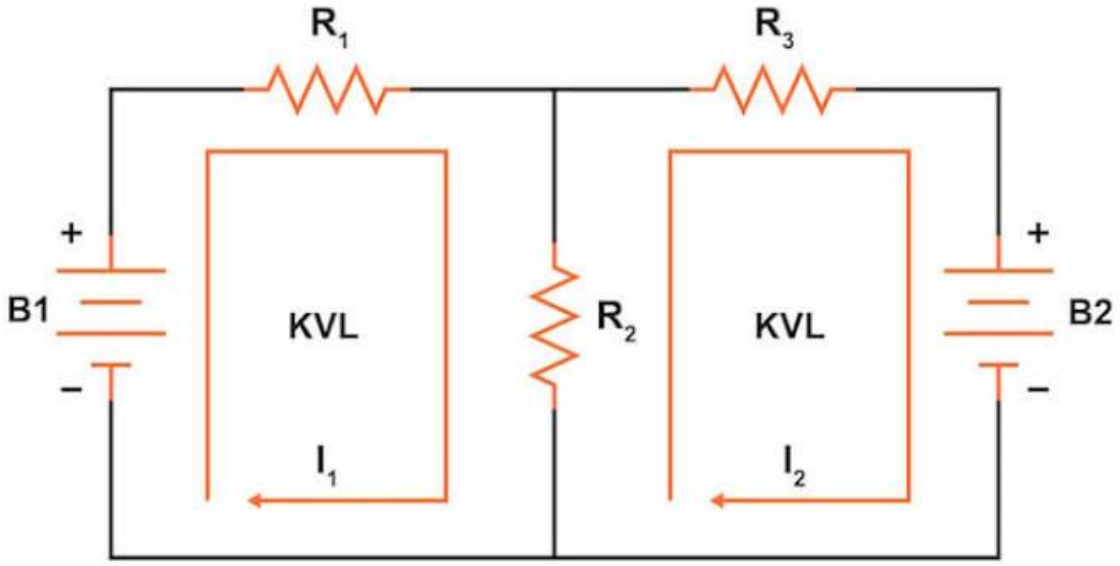
In simple words: Loop Current Analysis is a technique in which KVL equations are formed for each circuit loop and solved simultaneously to find the currents flowing in the loops.

Steps for Loop Analysis

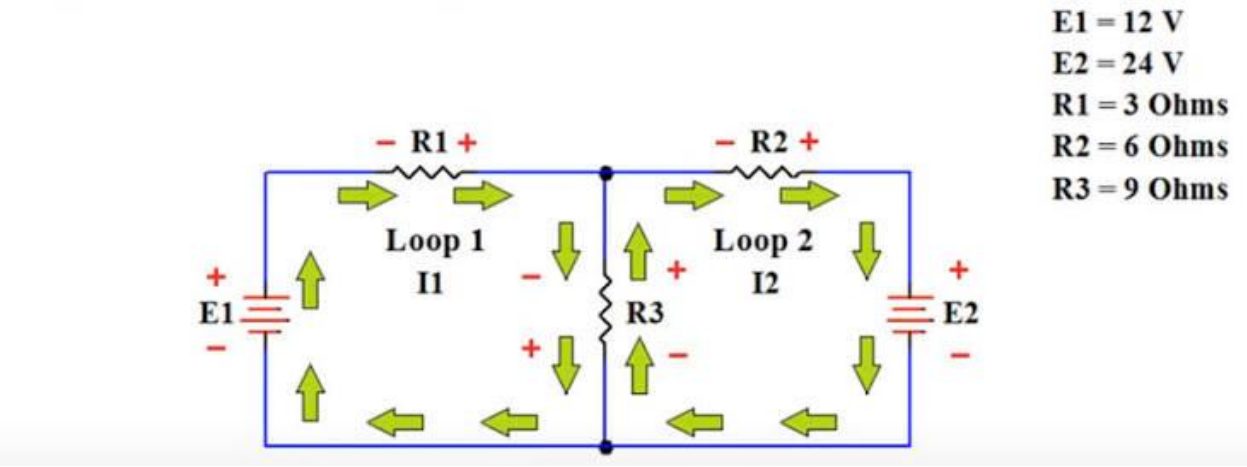
- **Identify loops: Find all independent, non-overlapping closed loops (meshes) in the circuit.**
- **Assign currents: Label a hypothetical current for each loop, usually pointing in the clockwise direction.**
- **Mark polarities: Set positive and negative voltage signs across resistors based on the entry and exit points of your assigned loop currents.**
- **Write KVL equations: Set the sum of all voltage drops and rises around each loop to equal zero using V = IR (Ohm's law).**
- **Handle shared components: For elements shared by two loops, calculate total current as the algebraic sum of the intersecting loop currents.**
- **Solve equations: Use substitution, elimination, or matrices to solve the system of linear equations for your unknown currents.**



Two loop network



Mess Current Method



How to perform Loop Current Analysis

2. Engineering Problem Set in Theory:

Q1) Define Error detection and correction codes.
Q2) Define Loop Current Analysis.
Q3) How to use Gauss seidel method in Large-Scale Power Flow & Mesh Networks.
Q4) Which method is used to identify the current loop analysis?
Q5) Define the application area for eigen values and eigen vectors.
Q6) In decoupling network equations, which mathematical tool is used?
Q7) Define the parity check in error detection codes.
Q8) During the Error Detection & Correction Codes, which mathematical tool is applied?

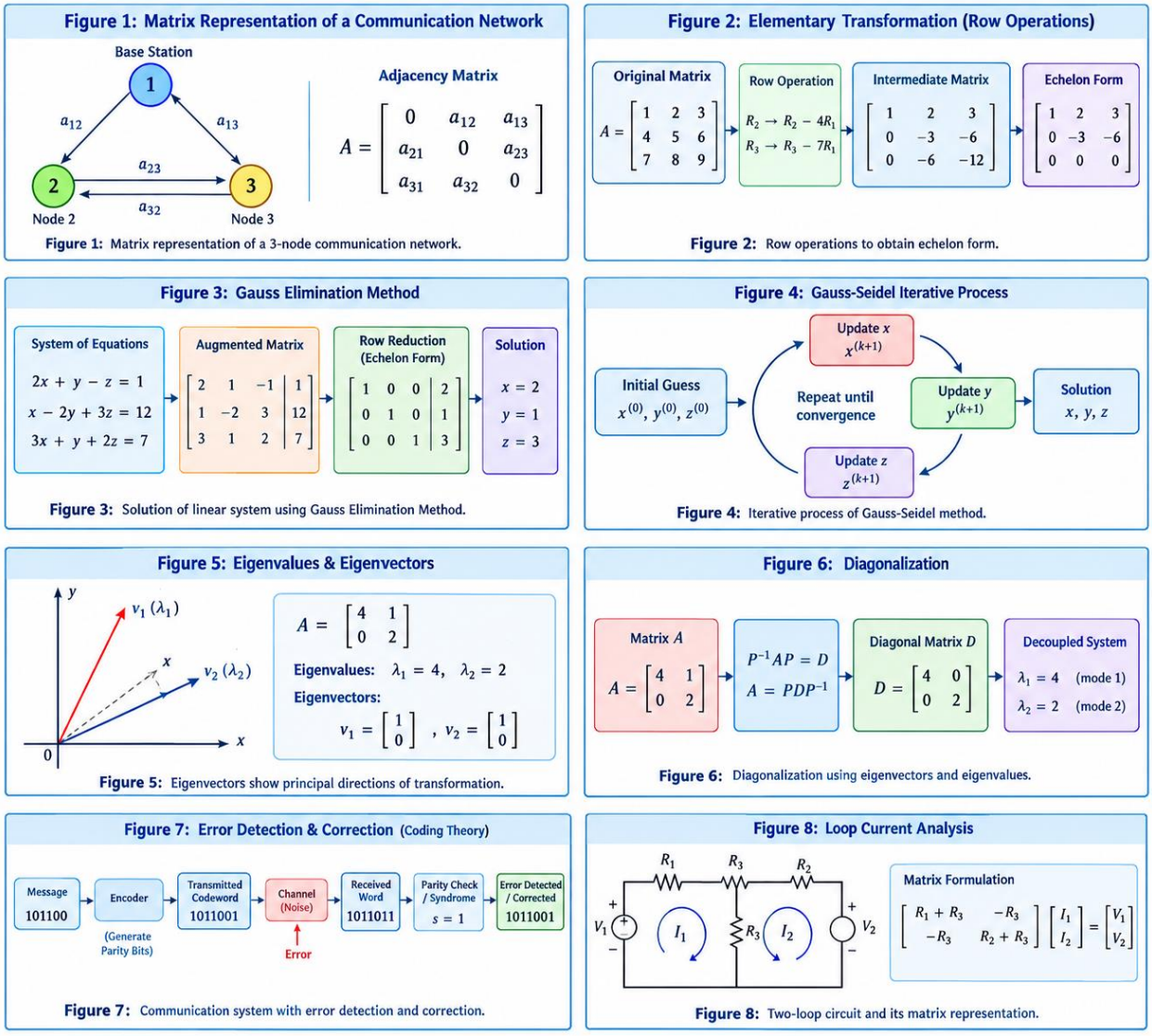


Figure Set: Engineering applications of Matrix Algebra—communication networks, row operations, Gauss elimination, Gauss-Seidel iteration, eigenvalues/eigenvectors, diagonalization, error detection/correction, and loop-current analysis.

3. Engineering Application Problem Set

The following 20 problems are arranged from fundamental matrix operations to engineering applications. Each solution shows the main computational steps so that the examples can be used directly for classroom teaching, practice sheets, and revision.

Problem 1: Matrix Representation of a Communication Network

Target Syllabus Topic: Introduction to Matrices

Engineering Application: Communication network connectivity

A communication network has three nodes. The directed link-count matrix is represented by A, where the entry a_ij denotes the number of direct links from node i to node j. Construct A for the links 1→2, 1→3, 2→3, 3→1, and 3→2. Also find A + A^T.

Problem 1 Solution:

Using the node order (1, 2, 3), place a 1 wherever a directed link exists and 0 otherwise.

$$A = [0\ 1\ 1;\ 0\ 0\ 1;\ 1\ 1\ 0]$$

The transpose represents the reverse-direction links.

$$A\text{^}T = [0\ 0\ 1;\ 1\ 0\ 1;\ 1\ 1\ 0]$$

$$A + A\text{^}T = [0\ 1\ 2;\ 1\ 0\ 2;\ 2\ 2\ 0]$$

Thus the matrix compactly represents the connectivity of the communication network.

Problem 2: Elementary Row Transformation

Target Syllabus Topic: Elementary Transformation

Engineering Application: System reduction in computational engineering

Reduce the matrix A = [[1,2,3],[2,5,8],[1,1,2]] using elementary row transformations and obtain an equivalent upper triangular matrix.

Problem 2 Solution:

$$A = [1\ 2\ 3;\ 2\ 5\ 8;\ 1\ 1\ 2]$$

Apply R2 → R2 − 2R1 and R3 → R3 − R1.

$$[1\ 2\ 3;\ 0\ 1\ 2;\ 0\ -1\ -1]$$

Now apply R3 → R3 + R2.

$$U = [1\ 2\ 3;\ 0\ 1\ 2;\ 0\ 0\ 1]$$

The matrix is in upper triangular form. Elementary transformations preserve the solution set of a corresponding homogeneous system.

Problem 3: Inverse of a Matrix by Elementary Transformations

Target Syllabus Topic: Inverse by E-transformation

Engineering Application: Circuit parameter recovery and inverse system models

Find the inverse of A = [[2,1],[3,2]] using elementary row transformations.

Problem 3 Solution:

$$[A \mid I] = [2\ 1 \mid 1\ 0;\ 3\ 2 \mid 0\ 1]$$

Apply R2 → 2R2 − 3R1.

$$[2\ 1 \mid 1\ 0;\ 0\ 1 \mid -3\ 2]$$

Apply R1 → R1 − R2.

$$[2\ 0 \mid 4\ -2;\ 0\ 1 \mid -3\ 2]$$

Apply R1 → (1/2)R1.

$$[1\ 0 \mid 2\ -1;\ 0\ 1 \mid -3\ 2]$$

$$A\text{^}(-1) = [2\ -1;\ -3\ 2]$$

Verification: AA^(-1) = I.

Problem 4: Rank by Echelon Form

Target Syllabus Topic: Rank of a Matrix by Echelon Form

Engineering Application: Detecting dependent measurements or redundant equations

Find the rank of A = [[1,2,3],[2,4,6],[1,1,2]] by reducing it to echelon form.

Problem 4 Solution:

$$A = [1\ 2\ 3;\ 2\ 4\ 6;\ 1\ 1\ 2]$$

Apply R2 → R2 − 2R1 and R3 → R3 − R1.

$$[1\ 2\ 3;\ 0\ 0\ 0;\ 0\ -1\ -1]$$

Interchange R2 and R3.

$$[1\ 2\ 3;\ 0\ -1\ -1;\ 0\ 0\ 0]$$

There are two non-zero rows.

$$rank(A) = 2$$

Hence the three rows are linearly dependent and the matrix does not have full rank.

Problem 5: Consistency of a Linear System Using Rank

Target Syllabus Topic: Rank of a Matrix by Echelon Form

Engineering Application: Solvability of coupled engineering equations

Determine whether the system x + y + z = 3, 2x + 2y + 2z = 6, x + y + 2z = 4 has a unique solution, infinitely many solutions, or no solution.

Problem 5 Solution:

$$[1\ 1\ 1 \mid 3;\ 2\ 2\ 2 \mid 6;\ 1\ 1\ 2 \mid 4]$$

Apply R2 → R2 − 2R1 and R3 → R3 − R1.

$$[1\ 1\ 1 \mid 3;\ 0\ 0\ 0 \mid 0;\ 0\ 0\ 1 \mid 1]$$

There are two independent equations in three unknowns.

$$rank(A) = rank(A|B) = 2 < 3$$

Therefore the system is consistent and has infinitely many solutions.

$$z = 1, \qquad x + y = 2$$

Let y = t. Then x = 2 − t, z = 1.

Problem 6: Gauss Elimination for Sensor Calibration

Target Syllabus Topic: Solution of System of Linear Equations by Gauss Elimination Method

Engineering Application: Sensor calibration and parameter estimation

Three calibration coefficients x, y and z satisfy x + y + z = 6, 2x + 3y + z = 11, and x + 2y + 3z = 14. Solve the system by Gauss elimination.

Problem 6 Solution:

$$[1\ 1\ 1 \mid 6;\ 2\ 3\ 1 \mid 11;\ 1\ 2\ 3 \mid 14]$$

R2 → R2 − 2R1; R3 → R3 − R1.

$$[1\ 1\ 1 \mid 6;\ 0\ 1\ -1 \mid -1;\ 0\ 1\ 2 \mid 8]$$

R3 → R3 − R2.

$$[1\ 1\ 1 \mid 6;\ 0\ 1\ -1 \mid -1;\ 0\ 0\ 3 \mid 9]$$

$$z = 3$$

$$y - z = -1 \Rightarrow y = 2$$

$$x + y + z = 6 \Rightarrow x = 1$$

$$(x, y, z) = (1,2,3)$$

Problem 7: Gauss Elimination for Loop Currents

Target Syllabus Topic: Solution of System of Linear Equations by Gauss Elimination Method

Engineering Application: Electrical loop-current analysis using Kirchhoff's voltage law

Three loop currents satisfy 10I1 − 2I2 = 8, −2I1 + 8I2 − 2I3 = 0, and −2I2 + 6I3 = 6. Find the loop currents using Gauss elimination.

Problem 7 Solution:

$$[10\ -2\ 0\ |\ 8;\ -2\ 8\ -2\ |\ 0;\ 0\ -2\ 6\ |\ 6]$$

Eliminating I1 and then I2 gives an upper triangular system.

$$10I1\ -\ 2I2\ =\ 8$$

$$7.6I2\ -\ 2I3\ =\ 1.6$$

$$-2I2\ +\ 6I3\ =\ 6$$

From the third equation, I2 = 3I3 − 3. Substituting into the second gives I3 = 1.

$$I2\ =\ 0$$

$$I1\ =\ 0.8$$

Therefore the loop-current solution is I1 = 0.8 A, I2 = 0 A, I3 = 1 A.

Problem 8: Gauss-Seidel Iterative Solution

Target Syllabus Topic: Solution of System of Linear Equations by Iterative Method (Gauss-Seidel Method)

Engineering Application: Iterative solution of large sparse engineering systems

Solve the system 10x − y + 2z = 6, −x + 11y − z = 25, 2x − y + 10z = −11 using the Gauss-Seidel method. Start with (0,0,0) and perform three iterations.

Problem 8 Solution:

Rewrite the equations in iterative form:

$$x\ =\ (6\ +\ y\ -\ 2z)/10$$

$$y\ =\ (25\ +\ x\ +\ z)/11$$

$$z\ =\ (-11\ -\ 2x\ +\ y)/10$$

Iteration 0: (x0,y0,z0) = (0,0,0).

$$x1 = 0.6$$

$$y1 = 2.3273$$

$$z1 = -1.2273$$

Iteration 2: use the newest available values.

$$x2 = 1.07818$$

$$y2 = 2.25917$$

$$z2 = -1.28772$$

Iteration 3:

$$x3 = 1.08346$$

$$y3 = 2.25325$$

$$z3 = -1.29137$$

The values are converging toward the solution (approximately 1.083, 2.252, −1.291).

Problem 9: Gauss-Seidel for a Thermal/Network Model

Target Syllabus Topic: Gauss-Seidel Method

Engineering Application: Iterative modeling in engineering computation

Solve approximately by Gauss-Seidel: 4x + y = 9 and x + 3y = 7, starting from (0,0). Perform four iterations.

Problem 9 Solution:

$$x = (9 - y)/4$$

$$y = (7 - x)/3$$

Iteration 1:

$$x1 = 2.25, \quad y1 = 1.5833$$

Iteration 2:

$$x2 = 1.8542, \quad y2 = 1.7153$$

Iteration 3:

$$x3 = 1.8212, \quad y3 = 1.7263$$

Iteration 4:

$$x4 = 1.8184, \quad y4 = 1.7272$$

Thus the iterative solution is approximately x = 1.818, y = 1.727.

Problem 10: Eigenvalues of a 2 × 2 System Matrix

Target Syllabus Topic: Eigen Values & Eigen Vectors

Engineering Application: Modal behavior of a coupled engineering system

Find the eigenvalues of A = [[4,1],[2,3]].

Problem 10 Solution:

$$|A - \lambda I| = |4-\lambda \ 1; \ 2 \ 3-\lambda| = 0$$

$$(4-\lambda)(3-\lambda) - 2 = 0$$

$$\lambda^2 - 7\lambda + 10 = 0$$

$$(\lambda - 5)(\lambda - 2) = 0$$

$$\lambda 1 = 5, \quad \lambda 2 = 2$$

Hence the system has two distinct eigenvalues.

Problem 11: Eigenvectors Corresponding to the Eigenvalues

Target Syllabus Topic: Eigen Values & Eigen Vectors

Engineering Application: Independent modes in signal and circuit systems

For A = [[4,1],[2,3]], find the eigenvectors corresponding to λ = 5 and λ = 2.

Problem 11 Solution:

For λ = 5:

$$(A - 5I)v = 0 \Rightarrow [-1 \ 1; \ 2 \ -2][x; y] = 0$$

$$y = x$$

$$v1 = [1; 1]$$

For λ = 2:

$$(A - 2I)v = 0 \Rightarrow [2\ 1;\ 2\ 1][x; y] = 0$$

$$2x + y = 0$$

$$v2 = [1; -2]$$

Thus the eigenpairs are (5,[1,1]^T) and (2,[1,-2]^T).

Problem 12: Eigenvalues of a 3 × 3 Engineering Matrix

Target Syllabus Topic: Eigen Values & Eigen Vectors

Engineering Application: Repeated modes in coupled systems

Find the eigenvalues of A = [[2,1,0],[1,2,0],[0,0,3]].

Problem 12 Solution:

$$|A - \lambda I| = |2 - \lambda\ 1\ 0;\ 1\ 2 - \lambda\ 0;\ 0\ 0\ 3 - \lambda|$$

$$= (3 - \lambda)[(2 - \lambda)\text{^}2 - 1]$$

$$= (3 - \lambda)(1 - \lambda)(3 - \lambda)$$

$$(\lambda - 1)(\lambda - 3)\text{^}2 = 0$$

$$\lambda = 1, 3, 3$$

Therefore λ = 3 is a repeated eigenvalue of algebraic multiplicity 2.

Problem 13: Diagonalization of a Matrix

Target Syllabus Topic: Diagonalization

Engineering Application: Decoupling coupled engineering systems

Diagonalize A = [[4,1],[2,3]] using its eigenvalues and eigenvectors.

Problem 13 Solution:

From Problems 10 and 11, the eigenvalues are 5 and 2 with eigenvectors [1,1]^T and [1,-2]^T.

$$P = [1\ 1;\ 1\ -2]$$

$$D = [5\ 0;\ 0\ 2]$$

Since the eigenvectors are linearly independent, P is invertible.

$$P\text{^}(-1) = (1/-3)[-2\ -1;\ -1\ 1] = [2/3\ 1/3;\ 1/3\ -1/3]$$

$$A = P\,D\,P\text{^}(-1)$$

Hence A is diagonalizable. The diagonal form allows the coupled system to be represented in independent modal coordinates.

Problem 14: Using Diagonalization to Compute a Matrix Power

Target Syllabus Topic: Diagonalization

Engineering Application: Repeated application of a system transformation

Using A = [[4,1],[2,3]] and its diagonalization, compute A^3.

Problem 14 Solution:

$$A = P\,D\,P\text{^}(-1) \Rightarrow A\text{^}3 = P\,D\text{^}3\,P\text{^}(-1)$$

$$D\text{^}3 = [125\ 0;\ 0\ 8]$$

Multiplying gives:

$$A\text{^}3 = [74\ 37;\ 74\ 37]$$

This illustrates why diagonalization is useful for repeated system updates and discrete-time models.

Problem 15: Parity-Check Matrix for Error Detection

Target Syllabus Topic: Applications: Error Detection and Correction Codes

Engineering Application: Digital communication error detection

A single-bit error occurs in a 3-bit received word. Use the parity-check matrix H = [[1,1,0],[0,1,1]] to determine whether the received word r = [1,0,1]^T contains an error.

Problem 15 Solution:

$$Hr = [1\ 1\ 0;\ 0\ 1\ 1][1;0;1]$$

$$= [1;1]$$

In binary arithmetic, the syndrome is non-zero.

$$s = [1;1] \neq [0;0]$$

Therefore an error is detected. The non-zero syndrome indicates that the received vector is not a valid codeword.

Problem 16: Error Location Using a Parity-Check Matrix

Target Syllabus Topic: Applications: Error Detection and Correction Codes

Engineering Application: Parity-based error detection in digital communication

Let H = [[1,0,1,1],[0,1,1,1]] and suppose the received word is r = [1,1,0,1]^T. Compute the syndrome and identify the error position.

Problem 16 Solution:

$$s = Hr$$

$$s = [1\ 0\ 1\ 1;\ 0\ 1\ 1\ 1][1; 1; 0; 1]$$

$$s = [0; 0]$$

The syndrome is zero, so the received word satisfies the parity checks and no error is detected.

Hence the received word is accepted as a valid codeword under this parity-check matrix.

Problem 17: Generator Matrix for a Linear Block Code

Target Syllabus Topic: Applications: Error Detection and Correction Codes

Engineering Application: Encoding information for digital communication

Let the generator matrix be G = [[1,0,1],[0,1,1]]. Encode the information words m1 = [1,0] and m2 = [1,1].

Problem 17 Solution:

For a linear block code, the codeword is c = mG (mod 2).

$$m1G = [1\ 0][1\ 0\ 1;\ 0\ 1\ 1] = [1\ 0\ 1]$$

$$m2G = [1\ 1][1\ 0\ 1;\ 0\ 1\ 1] = [1\ 1\ 0]\ (mod\ 2)$$

Therefore the encoded words are 101 and 110.

Problem 18: Loop Current Analysis of a Two-Loop Circuit

Target Syllabus Topic: Applications: Loop Current Analysis

Engineering Application: Electrical circuit design using Kirchhoff's laws

Two loop currents I1 and I2 satisfy 8I1 − 2I2 = 10 and −2I1 + 6I2 = 4. Solve for the loop currents using matrix algebra.

Problem 18 Solution:

$$[8\ -2;\ -2\ 6][I1;I2]\ =\ [10;4]$$

Using Gauss elimination, multiply the first equation by 1/4: 2I1 − 0.5I2 = 2.5.

Substitute into the second equation.

$$-2I1\ +\ 6I2\ =\ 4$$

From the first equation I1 = 1.25 + 0.25I2. Hence:

$$-2(1.25\ +\ 0.25I2)\ +\ 6I2\ =\ 4$$

$$5.5I2\ =\ 6.5$$

$$I2\ =\ 1.1818\ A$$

$$I1\ =\ 1.5455\ A$$

Thus the loop currents are approximately I1 = 1.546 A and I2 = 1.182 A.

Problem 19: Loop Current Analysis of a Three-Loop Network

Target Syllabus Topic: Applications: Loop Current Analysis

Engineering Application: Multi-loop electrical network analysis

Consider the matrix equation ZI = V for a three-loop network: [[10,-2,0],[-2,8,-1],[0,-1,6]][I1,I2,I3]^T = [12,0,6]^T. Find the loop currents.

Problem 19 Solution:

$$10I1\ -\ 2I2\ =\ 12$$

$$-2I1\ +\ 8I2\ -\ I3\ =\ 0$$

$$-I2\ +\ 6I3\ =\ 6$$

From the third equation, I2 = 6I3 − 6.

Substitute in the second equation:

$$-2I1\ +\ 8(6I3\ -\ 6)\ -\ I3\ =\ 0$$

$$2I1\ =\ 47I3\ -\ 48$$

Substitute this relation into the first equation. Solving gives:

$$I3\ \approx\ 1.565\ A$$

$$I2 \approx 3.391\ A$$

$$I1 \approx 1.878\ A$$

These currents satisfy the three Kirchhoff loop equations to rounding accuracy.

Problem 20: Combined Application: Eigenmodes of a Coupled Circuit/System

Target Syllabus Topic: Eigen Values, Eigen Vectors & Diagonalization

Engineering Application: Modal analysis of coupled electrical/signal systems

A simplified coupled engineering system has system matrix A = [[3,1],[1,3]]. Find its eigenvalues and eigenvectors, diagonalize A, and explain the engineering meaning of the result.

Problem 20 Solution:

$$|A - \lambda I| = |3-\lambda\ 1;\ 1\ 3-\lambda| = 0$$

$$(3-\lambda)\text{^}2 - 1 = 0$$

$$(\lambda-4)(\lambda-2) = 0$$

$$\lambda 1 = 4, \qquad \lambda 2 = 2$$

For λ1 = 4:

$$[-1\ 1;1\ -1][x;y] = 0 \Rightarrow y = x$$

$$v1 = [1;1]$$

For λ2 = 2:

$$[1\ 1;1\ 1][x;y] = 0 \Rightarrow y = -x$$

$$v2 = [1;-1]$$

Form the modal matrix and diagonal matrix:

$$P = [1\ 1;1\ -1],\ D = [4\ 0;0\ 2]$$

$$A = P\ D\ P\text{^}(-1)$$

Engineering interpretation: the original coupled variables can be expressed as two independent modes. The eigenvectors represent the mode shapes, while the eigenvalues represent the corresponding modal factors.

4. Summary of Engineering Applications

Application	Mathematical Role
Error detection and correction codes	Matrices represent generator and parity-check operations; the syndrome identifies invalid received words.
Loop Current Analysis	Kirchhoff equations are assembled into ZI = V and solved directly or iteratively.
Network/System Modeling	Matrices compactly represent connectivity, interactions, and coupled variables.
Modal Analysis	Eigenvalues and eigenvectors identify independent modes of coupled systems.
Diagonalization	A coupled transformation can be converted to independent modal coordinates, simplifying repeated computations.